PATIENT CARE THEORY 2

UNIT 12, PART 1: Abdominal Pain

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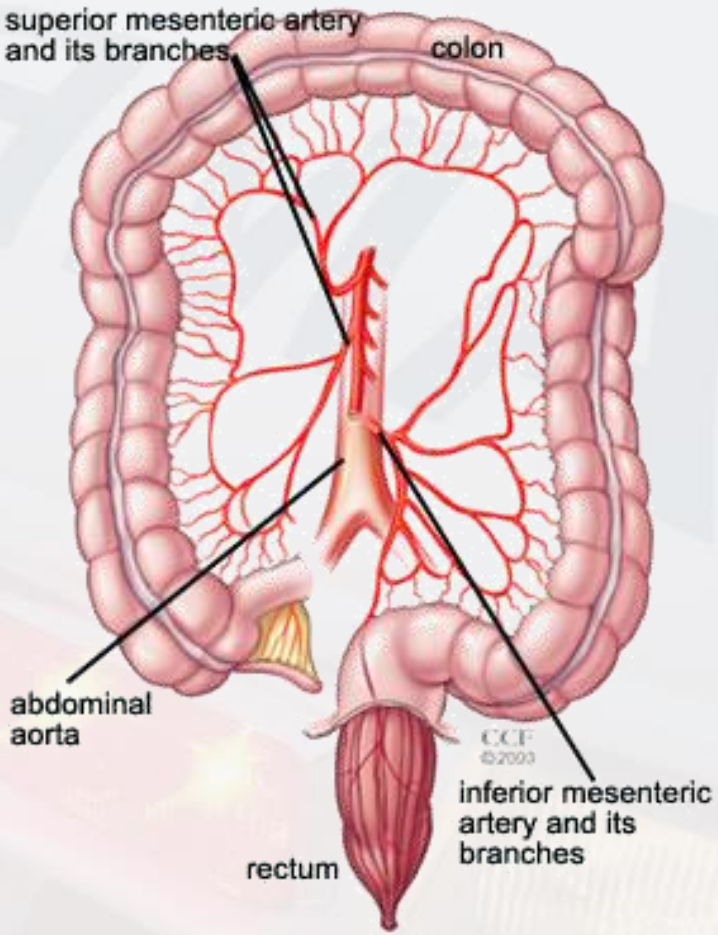
Learning Objectives

- Discuss the signs and symptoms and differentiate among local, general and peritoneal inflammations relative to acute abdominal pain
- Discuss the pathophysiology and management of acute gastrointestinal conditions

Causes of Abdominal Pain

Mesenteric artery occlusion

- found in people with heart conditions like atrial fibrillation, atrial flutter and those who have had heart valves replaced
- occlusion causes ischemia and pain out of proportion to their appearance
- mortality rate of 70 – 90%



Causes of Abdominal Pain

flare up after eating fatty, high cholesterol foods

Cholecystitis (inflammation of the gallbladder)

Cholelithiasis *(formation of gallstones)*

- *Cholecystitis* is caused by gallstones 90% of the time
  - Can also be caused by bacterial infections, tumors, bile duct abnormalities
- Stones form in the gallbladder from cholesterol saturated in bile which then calcifies
- *May obstruct the bile duct (biliary colic)*
- sharp pain localized in the RUQ, colicky in nature
- restless, nausea and vomiting
- pain can radiate into back, *chest and/or neck, or* irritate the diaphragm resulting in referred pain in the right shoulder
- **Murphy's sign** – pain on palpation under the right costal margin
- often have had past occurrences – usually requires surgical intervention to manage

comes and goes

under rib line

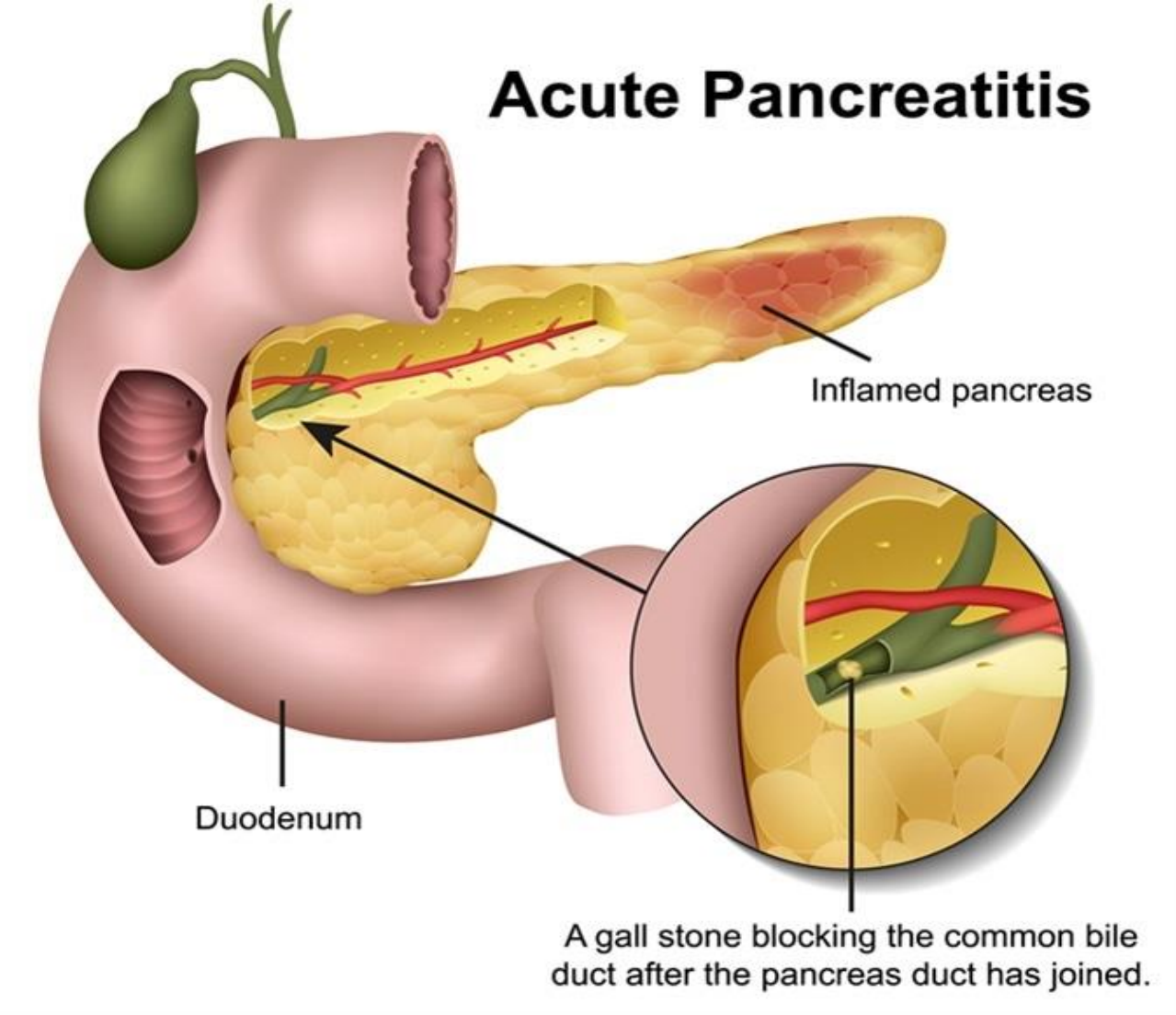
Murphy’s sign ☺

Causes of Abdominal Pain

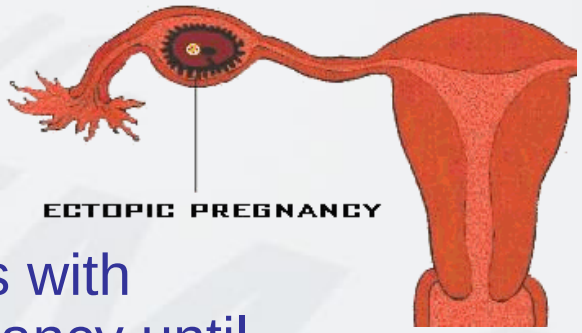
Pancreatitis

- inflammation of the pancreas due to activation of digestive enzymes while still in the pancreas
  - Consumption of alcohol, smoking
  - Obstructions of the common bile duct
  - Certain medications
- Overall mortality high (30-40%) (sepsis and shock)
- S&S include
  - sharp twisting pain in the LUQ or in the epigastric area ->radiating to the back
  - fever, nausea/vomiting (excessive uncontrolled vomiting)
  - abdominal distention
  - diaphoresis
  - Rebound tenderness ; bumps on the road
  - Fetal position

putting pressure may hurt or relieve pain, but letting go of pressure causes high pain


Acute Pancreatitis
Inflamed pancreas
Duodenum
A gall stone blocking the common bile duct after the pancreas duct has joined.

Causes of Abdominal Pain

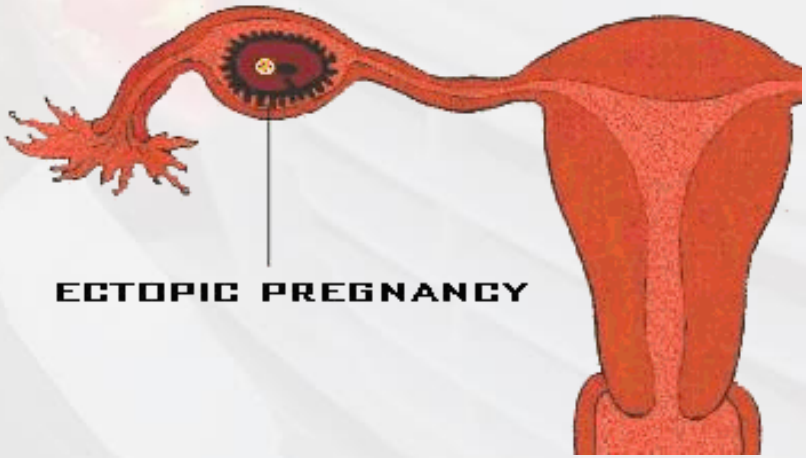


Ectopic Pregnancy

- assume females of child bearing years with abdominal pain have an ectopic pregnancy until proven otherwise
- fertilized egg implants outside the uterine cavity
- Presents with increasing lower quadrant abdominal pain increasing over time
- ***Rupture*** *- presents with sudden extreme unilateral lower pain low in the Abd – worsened by palpation*
- referred pain to shoulder
- Rupture usually occurs between 6 and 16 weeks
- has menstrual cycle been regular?
- any bleeding or other coloured vaginal fluid?

Causes of Abdominal Pain

Ectopic Pregnancy



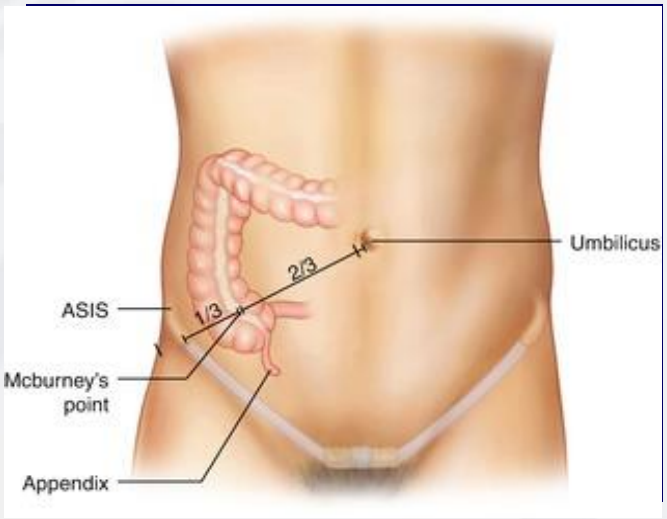
Last Menstrual Period (LMP): 6 weeks

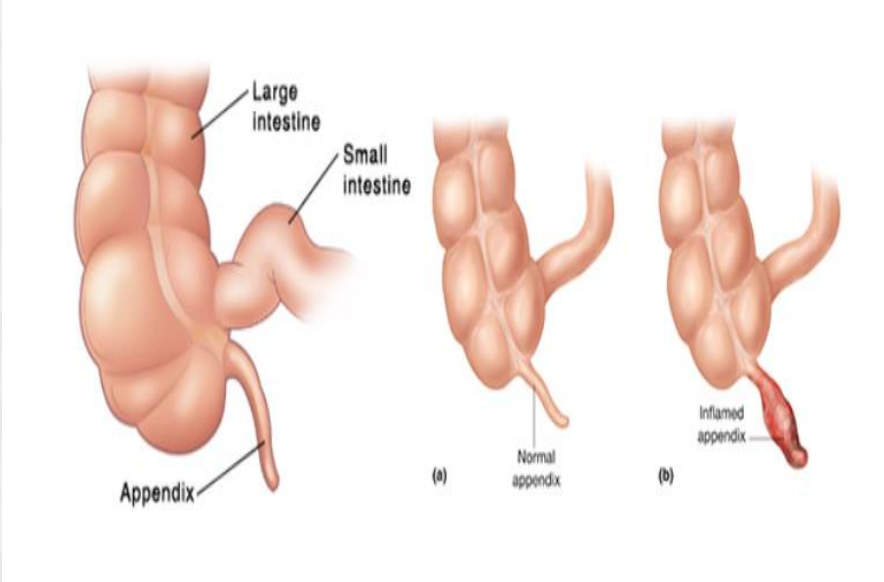
Causes of Abdominal Pain

dull, poorly localized. Somatic pain is musculoskeletal/skin pain, whic is more localzed

Appendicitis

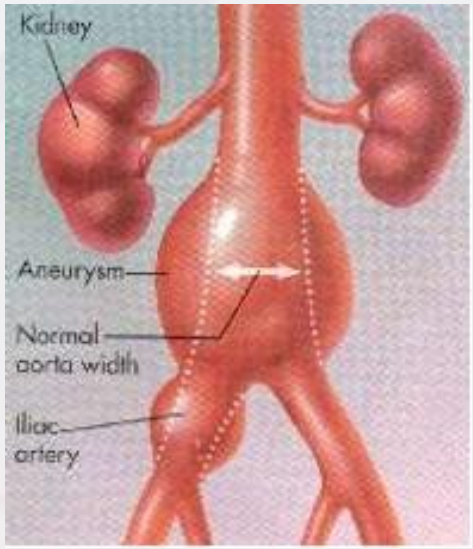
- most common abdominal emergency in older children and young adults
- caused by viral or bacterial infection or obstruction
- becomes gangrenous and/or ruptures
- early onset presents as visceral pain
- later presentation is somatic, intense and localized to “McBurney’s Point”
- Often misdiagnosed d/t wide variety of accompanying signs and symptoms





Causes of Abdominal Pain

- **Abdominal Aortic Aneurysm**
  - AAA
  - usually found between the bottom of the renal arteries and the top of the iliac arteries
  - Caused by atherosclerosis
  - affects more men then women
  - usually over 50 years old
  - S&S include:
    - tearing ripping pain through to the back *OR* unexplained low back pain
    - Reduced pulses in legs or diminished/absent peripherally (pedal pulse)
    - Numbness or tingling in the lower extremities
    - Orthostatic hypotension
  - look for a pulsating mass (lack of pulsating mass does not rule out AAA)



Causes of Abdominal Pain

Ketorolac is reccomended. it's more effective for organ/visceral pain. It can relax smooth muscles too (instead of skeletal).

Regular NSAIDs (acetameniphen and ibuprofen) more for somatic pains.

Renal Colic

- presents with sudden severe pain..
- flank pain -> may be radiating down to groin
- writhing in pain
- vomiting with the pain
- +/- dysuria
- may have a previous history of the same
  - Believed to have a genetic component
- blood in the urine..new or ongoing for a couple of days.

Causes of Abdominal Pain

Epigastric pain

- Gastro-esophageal reflux Disease or heart burn (GERD) is caused from the back up of stomach acid into the esophagus
- Peptic ulcers may be the cause. This is when there is a break in the lining of the esophagus or the stomach
  - NSAID use, Helicobacter pylori bacteria, Zollinger-Ellison Syndrome
- Gnawing or burning feeling
- REMEMBER this can be cardiac related

Causes of Abdominal Pain

- **UPPER GI TRACT DISEASES**
  - Esophageal Varices
  - Acute gastroenteritis
  - Chronic gastroenteritis
  - Peptic Ulcers
- **LOWER GI TRACT DISEASES**
  - Ulcerative Colitis
  - Diverticulitis
  - Hemorrhoids
  - Bowel Obstruction

varicose veins in lower esophagus

inflammation of small pouches (diverticulla) in colon

Causes of Abdominal Pain

Gastrointestinal Bleeding

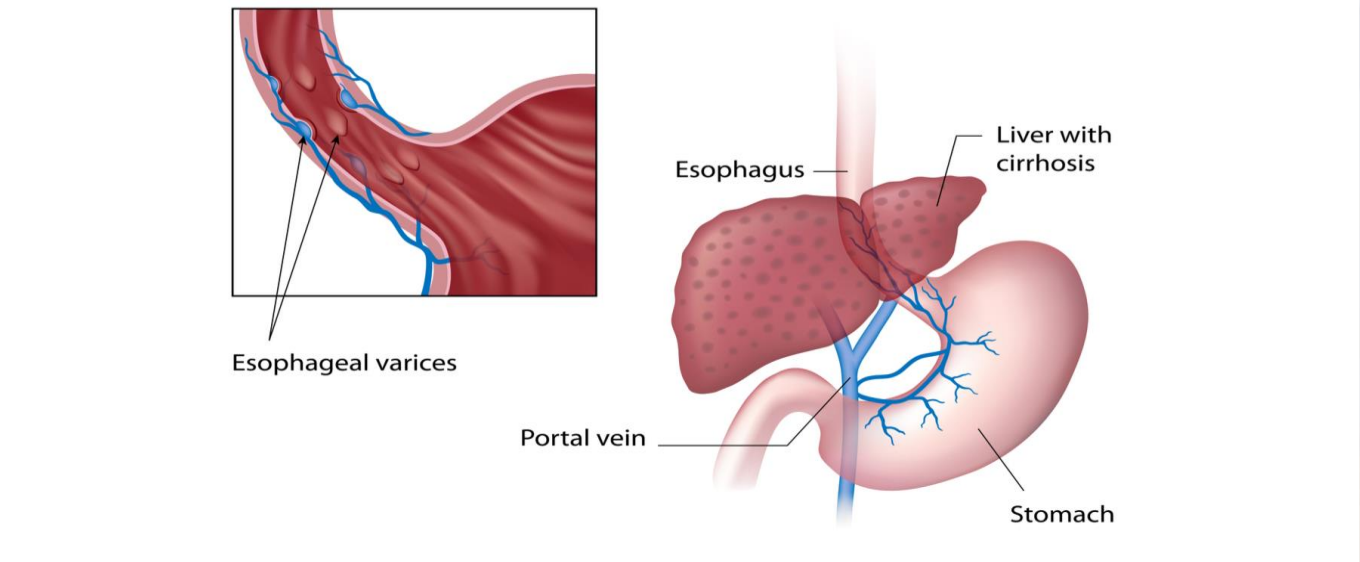
- Upper GI bleeds may be obvious or present slowly
- Blood is extremely irritating to the GI tract
  - This leads to nausea and vomiting in majority of upper GI bleeds (***hematemesis***)
  - Upper (usually)/Lower GI tract bleeding may present with ***melena*** (in order for melena to be seen, ~150mL of blood must be in the GI tract for 5-8 hours) OR:
  - ***Hematochezia –*** bright red (Frank) blood through the anus – usually in the stool

vomit appears to have red coffee grounds

Causes of Abdominal Pain

Gastrointestinal Bleeding

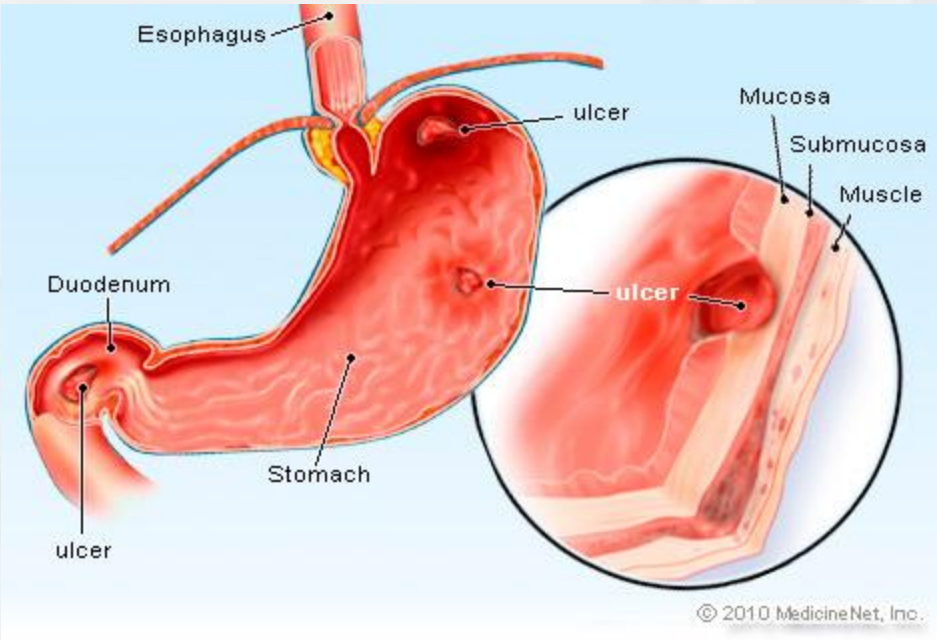
- ruptured esophageal varices



Causes of Abdominal Pain

Gastrointestinal Bleeding

- peptic ulcer disease (accounts for ~ 50% of upper GI bleeds)



Causes of Abdominal Pain

Gastrointestinal Bleeding

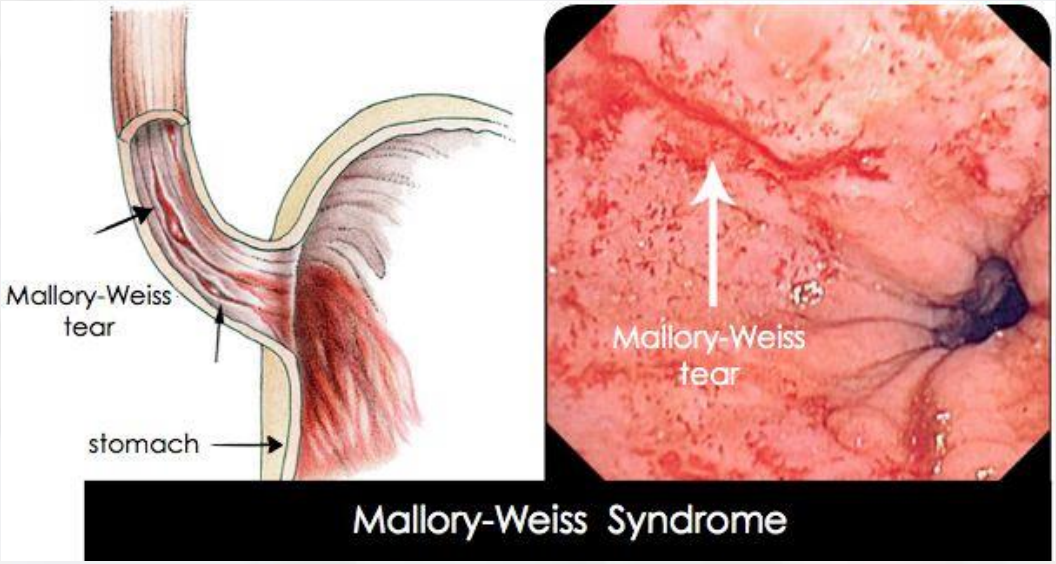
- Gastroenteritis – inflammation of the GI mucosa (stomach and intestines)
  - Acute – sudden onset
  - Chronic – long term mucosal changes with permanent damage (primarily due to microbial infection)
- Most cases of gastroenteritis are viral. Bacterial gastroenteritis tends to make patients significantly more ill

NSAIDs should be taken with food to prevent this

Causes of Abdominal Pain

Gastrointestinal Bleeding

- Mallory-Weis tear at the esophageal gastric junction from severe retching



Causes of Abdominal Pain

Gastrointestinal Bleeding

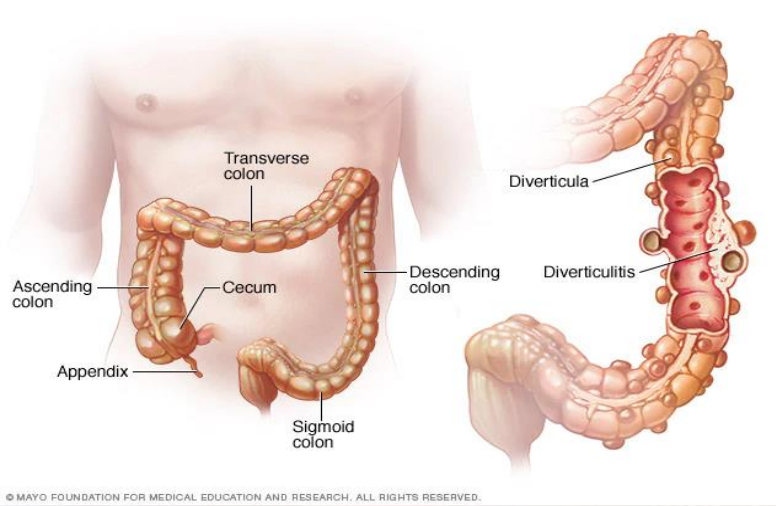
- Lower GI bleeds
  - ***Ulcerative Colitis*** – idiopathic inflammatory bowel disorder affecting the colon (large intestine)
  - Chronic ulcers in the mucosal layer
  - Common in the rectum and sigmoid colon, but can spread throughout the entire colon
  - Presents with colicky abdominal pain, N/V, sometimes a fever (if exacerbated by infection), weight loss
  - ***Crohn's disease*** - idiopathic inflammatory bowel disorder – similar to ulcerative colitis except it includes anywhere from the mouth to the rectum

Causes of Abdominal Pain

Gastrointestinal Bleeding

- Lower GI bleeds
  - ***Diverticulitis*** – inflammation of diverticula (complication of diverticulosis)
    - Diverticulosis – presence of diverticula +/- bleeding
    - Diverticula – small outpouchings in the mucosal lining of the intestinal tract

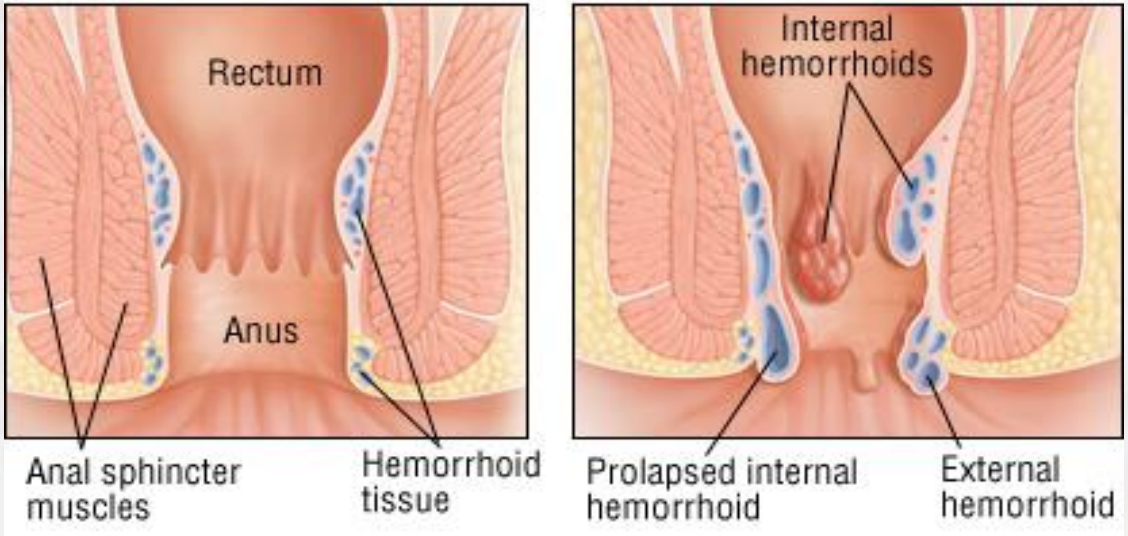
wearing away of mucosal lining can cause stool to enter diverticuli.
can lead to sepsis if burst



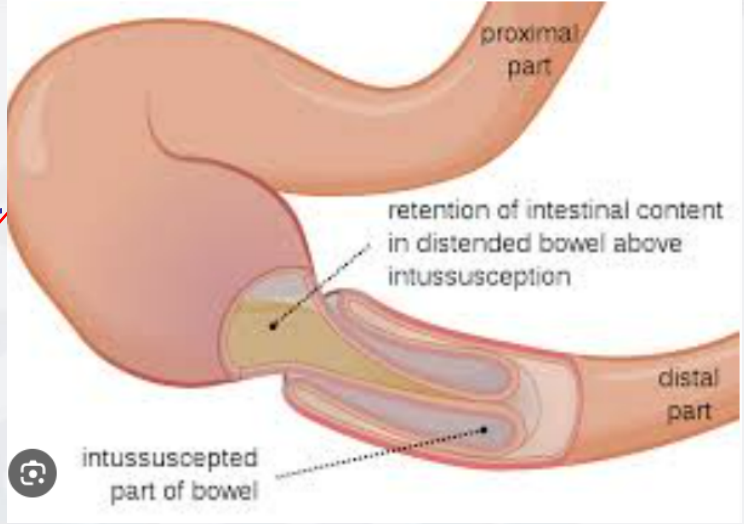
Causes of Abdominal Pain

Gastrointestinal Bleeding

- Lower GI bleeds
  - Hemorrhoids
    - Small masses of swollen veins in the anus or rectum



Causes of Abdomina



Bowel Obstruction

- common in the elderly
- intestinal pathway becomes blocked – fecal material – can be partial or complete
- ***Requires rapid diagnosis and treatment****
- Causes
  - **Hernias** – protrusion of an organ through its protective sheath
  - **Intussusception** – part of the intestine slips in the part just distal to itself
  - **Adhesions** – union of normally separate tissue surfaces by a fibrous band of new tissue (surgeries, healed tears, ulcers)
  - **Volvulus** – twisting of the intestine on itself
  - **Infarction** – area of dead tissue caused by lack of blood

Causes of Abdominal Pain

Bowel Obstruction

- Obtain Hx of bowel movements, flatus, constipation (ie, no gas or bowel movement)
- Typical S&S abdominal distention, N/V crampy abdominal pain, hemodynamic instability
- Diffuse visceral pain
- Abrupt onset = likely an obstruction
- Look for scars from previous surgeries, trapped air in the abdominal tissue
- Significantly reduced bowel sounds

QUESTIONS?